CHAPTER 6

Bitcoin Network and Payments

Mastering Blockchain · 2nd Edition

TC789 — Teknologi Blockchain
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Learning Objectives

By the end of this lecture, you will be able to:

- Describe the Bitcoin peer-to-peer network architecture and node types
- Explain the node discovery and block synchronization protocols
- Understand protocol messages exchanged between Bitcoin nodes
- Identify and compare different types of Bitcoin wallets
- Explain how Bitcoin payments work and the role of BIPs
- Discuss advanced protocols: SegWit, Bitcoin Cash, Lightning Network

Chapter Roadmap

1

The Bitcoin Network

Node types & protocol messages

2

Node Discovery & Sync

How nodes find each other & sync blocks

3

Bitcoin Wallets

Types, security, and key management

4

Bitcoin Payments

POS, BIPs, Lightning Network

5

Bitcoin Innovation

SegWit, BCH, BTG, BIPs

The Bitcoin Network

A peer-to-peer network where nodes exchange transactions and blocks.

Full Node	Wallet + Miner + Full Blockchain + Network Routing. Most secure.
SPV Node	Simplified Payment Verification - Lightweight client: Wallet + Network Routing only. Downloads only block headers.
Solo Miner Node	Mining + Full Blockchain + Network Routing. No wallet.
Pool Protocol Server	Uses Stratum protocol (JSON-RPC over TCP). Used by mining pools.
Mining Node	Computes hashes only; submits via Stratum to pool servers.

Network	Magic value	Hex
main	0xD9B4BEF9	F9 BE B4 D9
testnet3	0x0709110B	0B 11 09 07

Bitcoin Protocol Messages

27 total message types. Key messages used in everyday network operation:

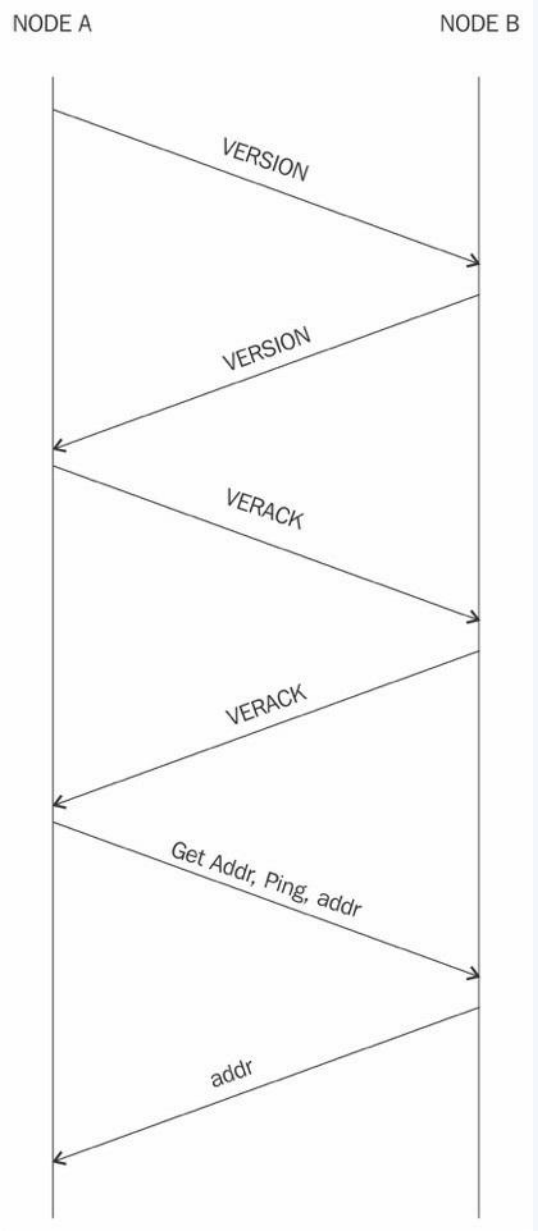
version / verack	Handshake — node introduces itself; peer accepts connection
inv	Advertise knowledge of blocks and transactions
getdata	Request a specific block or transaction by hash
getblocks	Request list of blocks after a known hash (up to 500)
getheaders / headers	Request / return block headers (up to 2,000)
tx	Send a transaction in response to getdata
block	Send a block in response to getdata
getaddr / addr	Request / provide list of known peer addresses

Default port: TCP 8333 (mainnet) · TCP 18333 (testnet)

Node Discovery Protocol

How Nodes Find Each Other

- On startup, client queries hardcoded DNS seeds
- DNS lookup returns peer IP addresses
- Client sends version → peer replies version + verack
- Handshake complete; peers exchange getaddr / addr
- Additional peers discovered iteratively
- Nodes send ping to keep connections alive

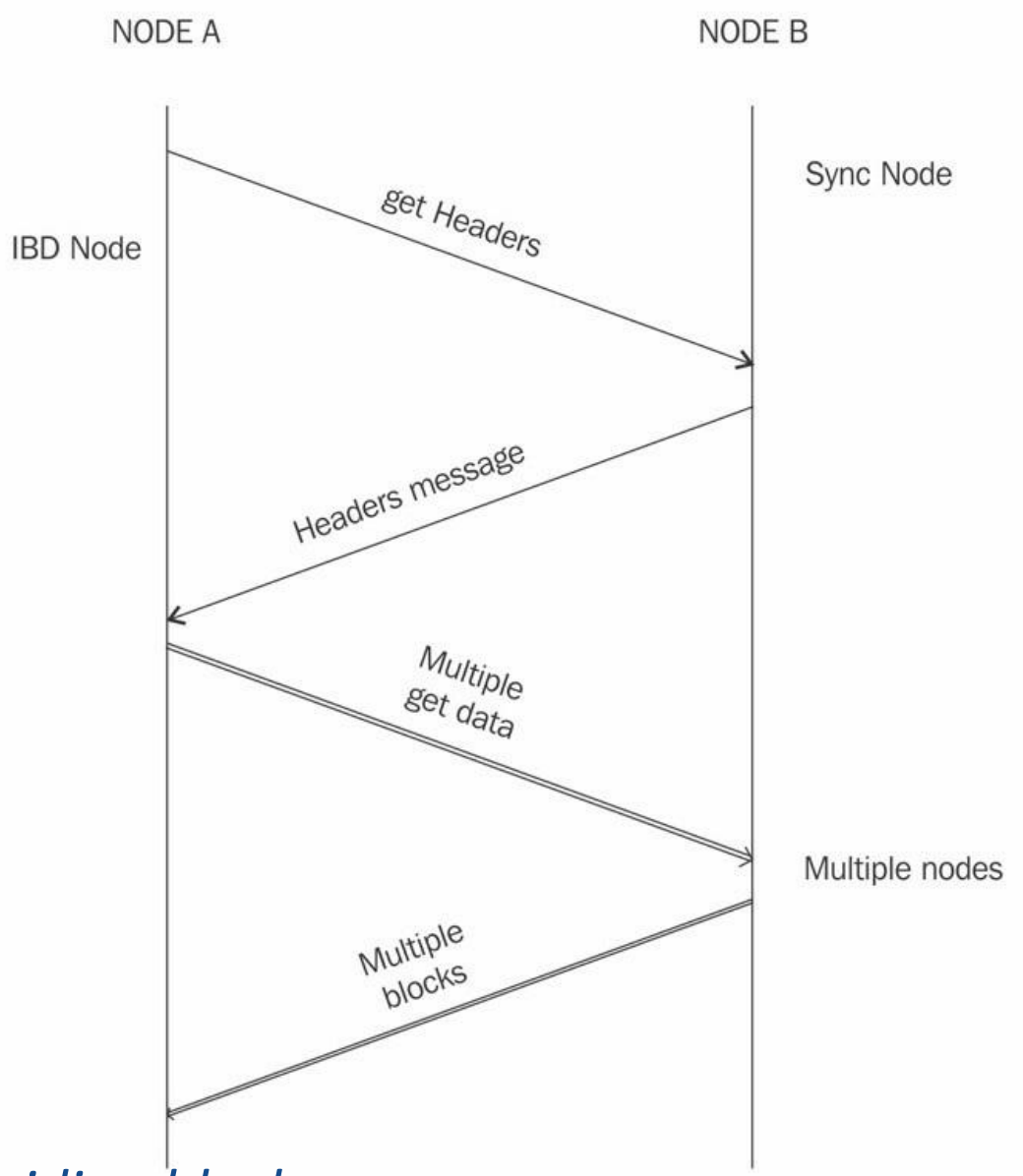


Example DNS seeds: seed.bitcoin.sipa.be · dnsseed.bluematt.me

Block Synchronization: Headers-First (v0.10.0+)

Initial Block Download (IBD)

- Blocks-first (pre-0.10): slow; downloaded full blocks sequentially
- Headers-first (v0.10.0+, Feb 2015): get headers first, then blocks in parallel
- IBD Node requests headers via getheaders from Sync Node
- Sync Node returns up to 2,000 block headers
- IBD Node downloads full blocks in parallel from multiple peers
- Sync time: days → hours

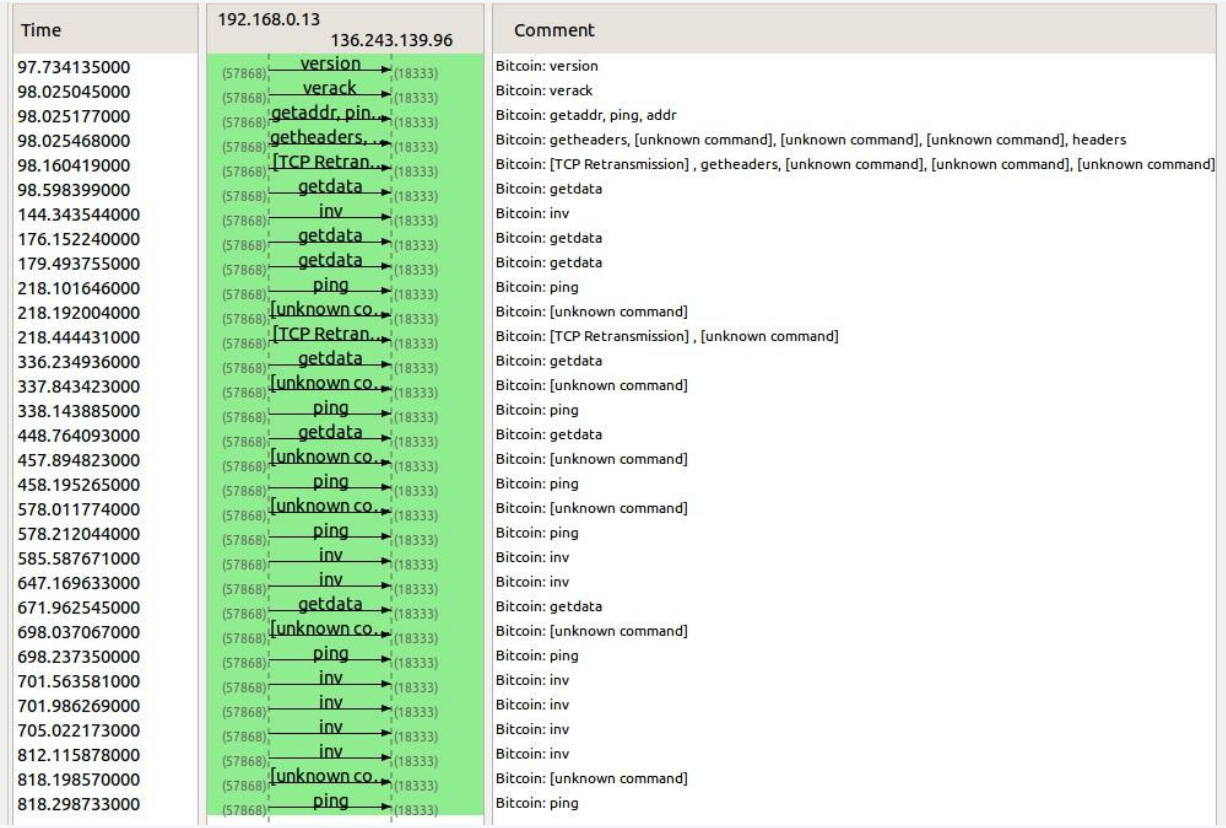


IBD Node = syncing node | Sync Node = peer providing blocks

Analyzing Bitcoin Protocol with Wireshark

Protocol Inspection & Learning

- Wireshark is a network packet analyzer
- Bitcoin dissector identifies protocol messages
- Useful for studying version, verack, getaddr, inv, addr messages
- Recommended: experiment on Bitcoin testnet (not mainnet)
- Shows command name, payload, and message flow in real-time



Testnet: https://en.bitcoin.it/wiki/Testnet | Wireshark: https://www.wireshark.org

SPV Clients & Bloom Filters

SPV Nodes (Lightweight Clients)

- Download only block headers, not full blockchain
- Verify payments using Merkle branches
- Request transactions from full nodes when needed
- Lower bandwidth; suitable for mobile/lightweight clients

Bloom Filters (BIP - Bitcoin Improvement Proposals 37)

- Probabilistic data structure (bit vector)
- False positives allowed — no false negatives
- SPV nodes use bloom filters to request only relevant transactions
- Introduced in BIP 37
- New messages: filterload, filteradd, filterclear
- Trade-off: strict = precise but less private; loose = more private

Privacy trade-off: Strict filter = your address may be inferred by peers. Loose filter = better privacy, higher bandwidth.

Bitcoin Wallets — Overview

A wallet is software (or hardware) that stores private keys — NOT coins.
Coins exist as UTXOs on the blockchain; wallets hold the keys to access them.

Key Concepts

- Private keys: 256-bit random numbers generated by wallet software
- Used to sign outgoing transactions (proof of ownership)
- Public key and Bitcoin address are derived from the private key
- Wallet.dat stored as Berkeley DB file in Bitcoin Core
- Losing your private key = losing access to your bitcoin permanently
- Wallets do NOT store balances — balance is calculated from UTXOs on-chain

Types of Bitcoin Wallets

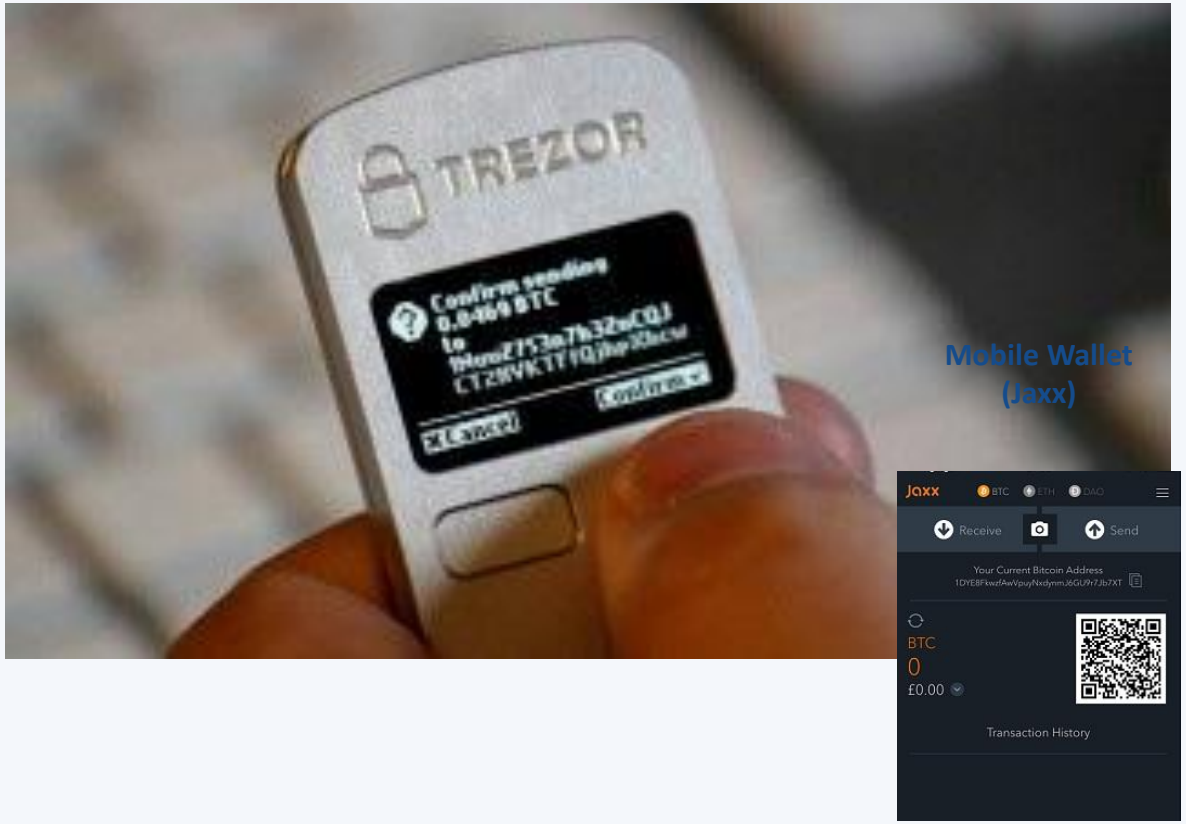
Type	Description	Example
Non-Deterministic	Random keys; hard to back up	Bitcoin Core (early)
Deterministic	Keys from seed via hash (BIP 39 mnemonics)	Easy recovery with 12-24 words
HD (Hierarchical Deterministic)	Tree of keys from one master seed (BIP32/44)	Trezor, Electrum, Jaxx
Brain Wallet	Key from memorized passphrase; brute-force risk	⚠ Not recommended
Paper Wallet	Printed key material; needs physical security	bitcoinpaperwallet.com
Hardware Wallet	Tamper-resistant device; most secure	Trezor, Ledger
Online Wallet	Cloud-hosted; requires trust in provider	GreenAddress
Mobile Wallet	Smartphone app; QR code scanning payments	Blockchain, Copay, Jaxx

Hierarchical Deterministic (HD) Wallets

How HD Wallets Work

- Defined in BIP 32 (structure) and BIP 44 (multi-account hierarchy)
- One master seed → generates master private key
- Master key → child keys → grandchild keys (tree structure)
- Full key hierarchy recoverable from just the master key
- BIP 39: human-readable mnemonic (12–24 words) represents the seed
- Highly portable — transfer wallet by sharing the seed phrase
- Examples: Trezor, Ledger, Jaxx, Electrum

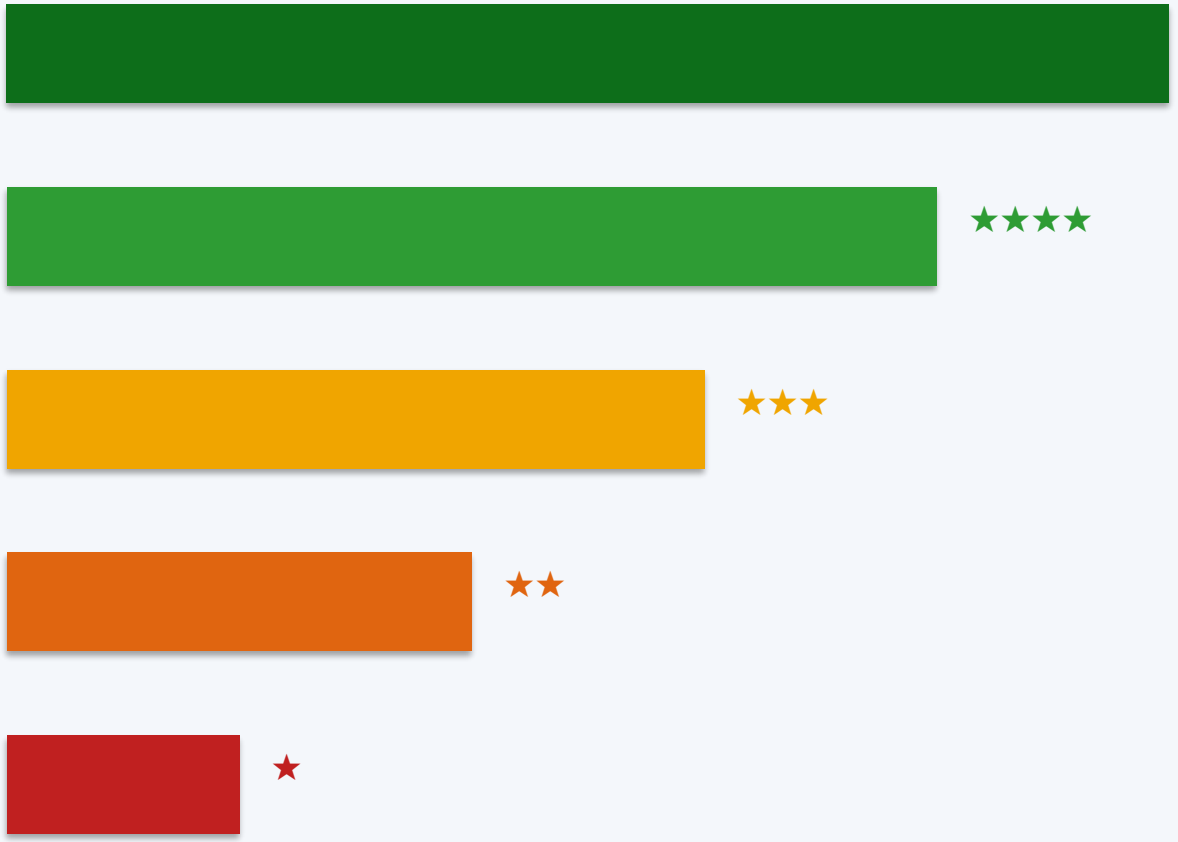
Trezor Hardware Wallet



Wallet Security Comparison

Security Level (Most → Least Secure)

Hardware Wallet	Tamper-resistant chip; keys never leave the device	★★★★★
Paper Wallet	Air-gapped; requires physical security (fire/theft risk)	★★★★
Mobile Wallet	Convenient; QR code payments; balanced security	★★★
Desktop Wallet	More exposed to malware; requires good OS hygiene	★★
Online Wallet	Hosted by third party; requires full trust in provider; highest risk	★



Bitcoin Payments

Step 1
Cashier enters fiat amount into POS terminal

Step 2
System generates QR code with Bitcoin URI (BIP 21)

Step 3
Customer scans QR code with mobile wallet

Step 4
Payment sent to seller's Bitcoin address

Step 5
Confirmation received; receipt printed

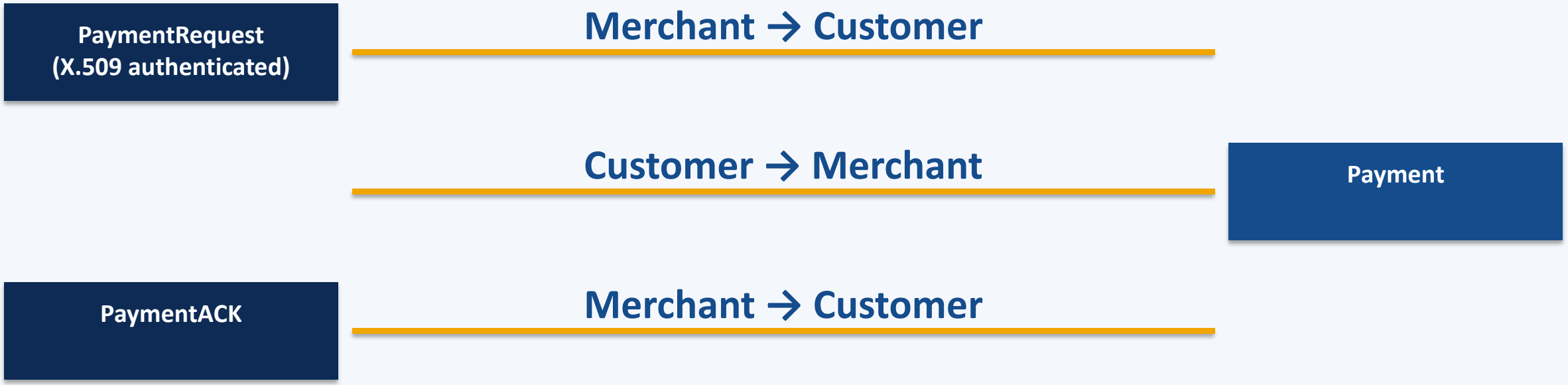
Payment Methods & Tools

- Bitcoin URI (BIP 21): bitcoin:ADDRESS?amount=X&label=Y
- QR codes: near universal wallet support
- POS terminals: XBTerminal, 34 Bytes POS
- Online payment processors: BitPay, Bitcoin Merchant Solutions
- Lightning Network: off-chain instant micropayments (BIP 2016)



Payment Protocol — BIP 70

BIP 70 defines a secure communication protocol between merchants and customers.

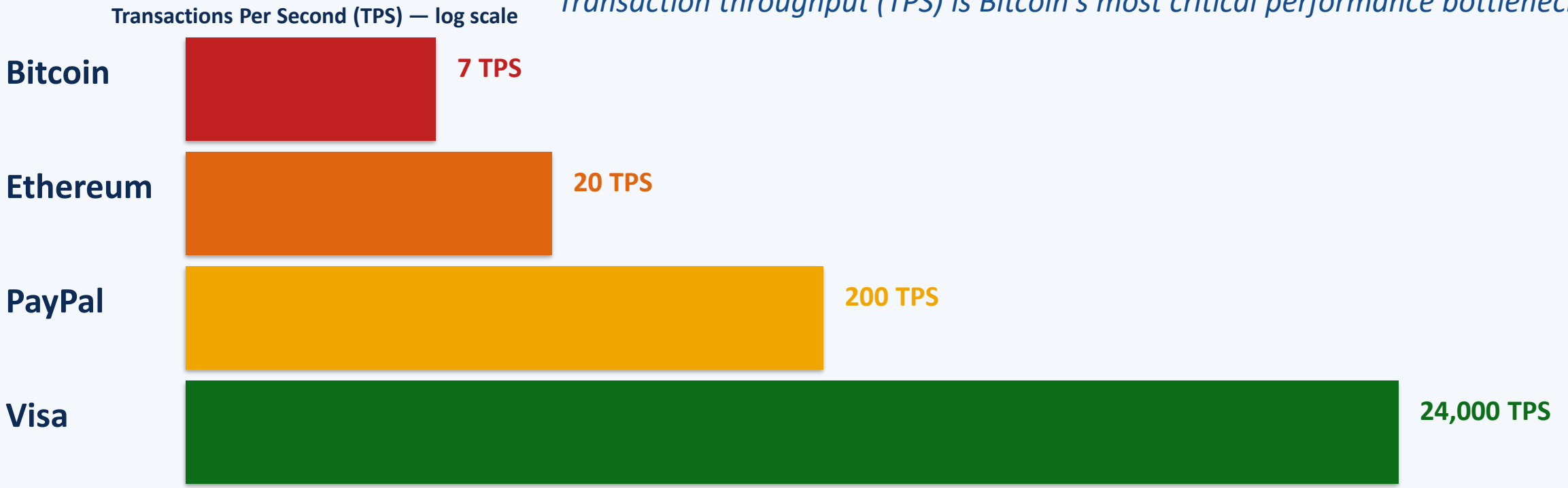


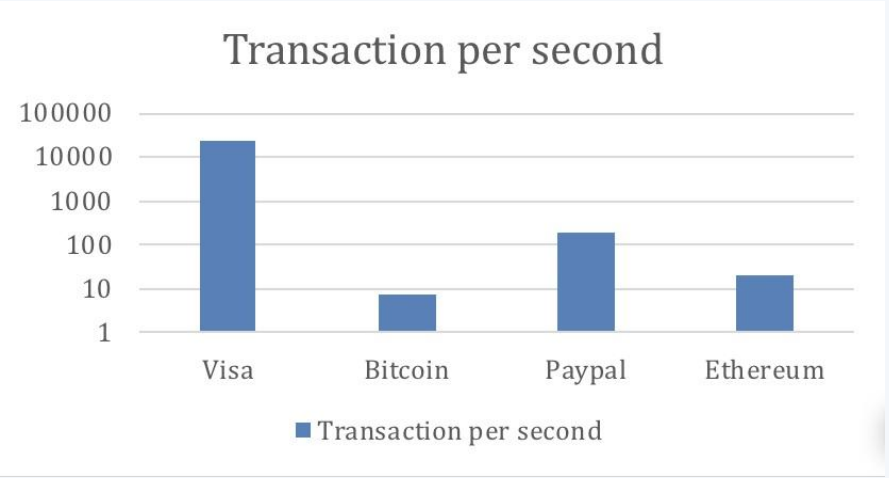
Key Features

- Uses X.509 certificates for merchant authentication (like HTTPS)
- Defends against man-in-the-middle attacks
- Provides secure proof of payment
- Supported by BIP 71 (MIME types) and BIP 72 (URI extensions)

Bitcoin's Scalability Challenge

Transaction throughput (TPS) is Bitcoin's most critical performance bottleneck.





Bitcoin processes ~3–7 TPS. Visa processes ~24,000 TPS.

Bitcoin Improvement Proposals (BIPs)

Documents used to propose, inform, or record changes to the Bitcoin ecosystem.

Standard BIP

Major protocol changes (block size, network protocol, tx verification)
Requires broad community consensus

Process BIP

Changes outside core protocol (governance, process improvements)
Implemented after community consensus

Informational BIP

Advisory or informational (design issues, ecosystem notes)
Does not require implementation

Notable BIPs: BIP 21 (URI) · BIP 32 (HD wallets) · BIP 37 (Bloom filters) · BIP 39 (Mnemonic) · BIP 70 (Payment Protocol) · BIP 141 (SegWit)

Segregated Witness (SegWit)

Soft fork activated August 24, 2017 · BIP 141, 143, 144, 145

Core Idea: Separate signature (witness) data from transaction data → smaller transactions → higher effective block capacity

Benefits

- Fixes transaction malleability (tx ID no longer based on signature)
- Cheaper fees: smaller transaction sizes
- Faster signing and verification
- Script versioning: future upgrades without hard fork
- Block weight limit: effective capacity ~1.6–2 MB (vs. 1 MB hard limit)

Technical Details

- Pre-SegWit: signature inside locking script
- Post-SegWit: signature stored separately (witness)
- Transaction types:
  - P2WPKH (Pay to Witness Public Key Hash)
  - P2WSH (Pay to Witness Script Hash)
- Backward compatible (soft fork — old clients still work)

Advanced Bitcoin Protocols

Bitcoin Cash (BCH)

Hard fork · 8 MB blocks
PoW / ASIC mining
Block interval: 10 sec – 2 hrs
Replay + wipe-out protection

Bitcoin Unlimited (BU)

No hard block size limit
Miners vote on block cap
Extreme thin blocks (bloom filter-based)
Parallel block validation

Bitcoin Gold (BTG)

Hard fork at block 491,407
Equihash algorithm (GPU-friendly)
ASIC-resistant
Restores decentralized mining

Note: All three proposals address Bitcoin's scalability and/or centralization issues via different approaches.

Lightning Network

Introduced 2016 · Off-chain payment solution · lightning.network

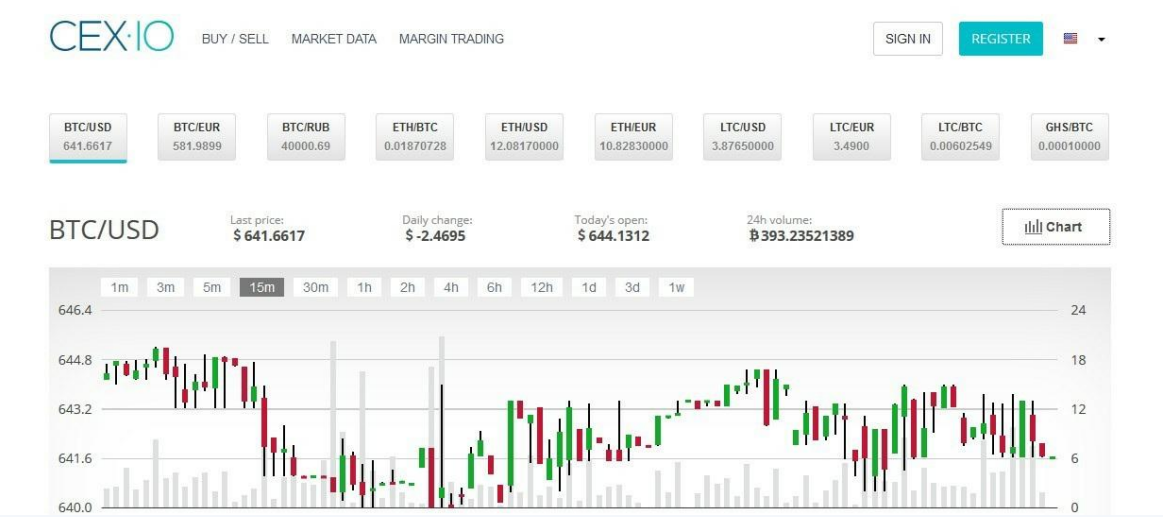
How It Works

- Payment channels run off the main Bitcoin blockchain
- Enables instant, low-fee micropayments
- Parties open a channel by locking funds in a multi-signature transaction
- Unlimited transactions within the channel; only open/close are on-chain
- Channels can be routed through the network (multi-hop payments)
- Dramatically increases Bitcoin's effective throughput
- Closes the gap between Bitcoin (~7 TPS) and traditional systems

Bitcoin Investment & Trading

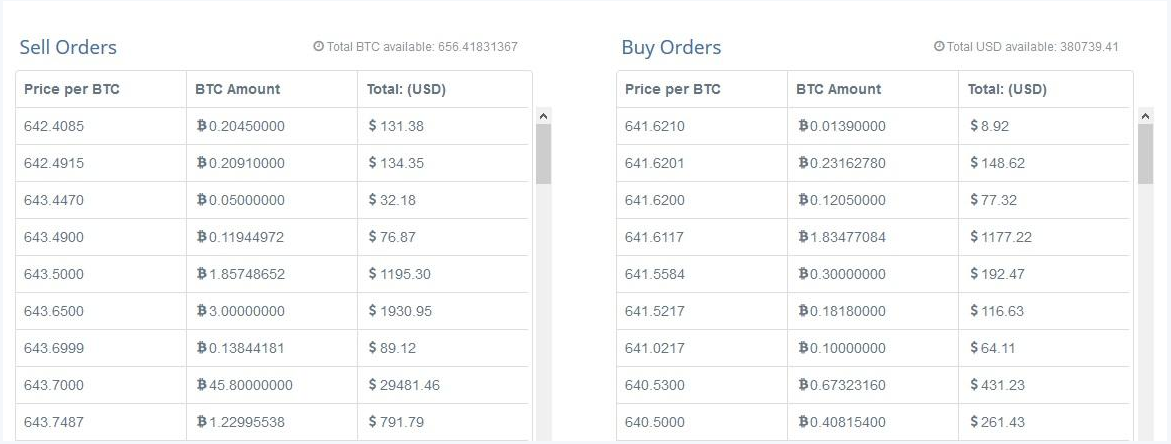
How Bitcoin is Traded

- Bitcoins traded on online exchanges (e.g., CEX.io, Binance)
- Trading options: spot, CFDs, spread betting, margin trading
- Long position: profit when price rises
- Short position: profit when price falls
- Can exchange between other cryptocurrencies



Order Types

- Order Book: lists all buy (bid) and sell (ask) orders
- Market Order: filled immediately when prices match
- Limit Order: filled at specified price or better
- Bid price: what buyer is willing to pay
- Ask price: what seller is willing to accept
- Trade occurs when bid = ask



Key Terms & Definitions

Full Node	A Bitcoin node that stores the complete blockchain and validates all transactions
SPV Node	Simplified Payment Verification — verifies payments using Merkle proofs without full blockchain
UTXO	Unspent Transaction Output — the 'coins' stored on the blockchain
Bloom Filter	Probabilistic data structure to test set membership with false positives, no false negatives
HD Wallet	Hierarchical Deterministic wallet — derives all keys from a single master seed (BIP32/44)
BIP	Bitcoin Improvement Proposal — formal document for proposing protocol changes
SegWit	Segregated Witness — separates signature data from transaction data (BIP141)
Lightning Network	Layer-2 payment channel network enabling off-chain, instant Bitcoin micropayments
Mnemonic Seed	12–24 human-readable words representing an HD wallet seed (BIP39)
Stratum Protocol	Line-based, JSON-RPC over TCP protocol used by mining pools

Bitcoin Network — Benefits vs. Limitations

✓ Benefits

- ✓ Decentralized — no single point of failure
- ✓ Censorship-resistant global payments
- ✓ SPV clients allow lightweight participation
- ✓ HD wallets: easy backup and portability
- ✓ Open payment standards via BIPs
- ✓ Constant innovation (SegWit, Lightning Network)

✗ Limitations

- ✗ Low throughput (~3–7 TPS vs. Visa's 24,000)
- ✗ Initial blockchain sync takes several hours
- ✗ Privacy concerns with bloom filters
- ✗ Brain wallets vulnerable to brute force
- ✗ Online wallets require trust in third parties
- ✗ No legal tender status in most jurisdictions

Key Takeaways

1. Bitcoin is a peer-to-peer network with different node types serving different roles (full, SPV, miner, pool).
2. The headers-first sync (v0.10.0+) dramatically improved initial blockchain download — from days to hours.
3. Bloom filters let SPV clients verify transactions efficiently, with a privacy vs. bandwidth trade-off.
4. Wallets store private keys, not coins. Coins exist as UTXOs; losing your key = losing your bitcoin.
5. HD wallets (BIP32/44) with mnemonic seeds (BIP39) are the modern standard for key management.
6. Bitcoin payments use QR codes, Bitcoin URIs (BIP21), and secure payment protocols (BIP70).
7. SegWit (2017) is Bitcoin's most significant upgrade: fixes malleability, reduces fees, increases throughput.
8. Scalability (~7 TPS vs 24,000 TPS Visa) drives innovation: Lightning Network, BCH, BTG, BU.

Discussion & Q&A

1. Is Bitcoin truly decentralized if a few mining pools control most hash power?
2. Would you trust a hardware wallet more than an online wallet? Why?
3. Can Bitcoin realistically scale to Visa-level throughput? How?
4. What are the risks and benefits of a small business accepting Bitcoin?
5. How does SegWit differ from a hard fork like Bitcoin Cash?